DEBT SUSTAINABILITY I

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Piotr Żoch

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PLAN

- We review several approaches to debt sustainability analysis.
- Can the government service its current outstanding debt? Or will a fiscal consolidation be needed?
- Is the outstanding public debt and its projected path consistent with those of the government's revenues and expenditures?

CLASSIC DEBT SUSTAINABILITY ANALYSIS

GOVERNMENT BUDGET CONSTRAINT

- The government flow budget constraint:

$$B_{t+1} = R_{t+1} B_t + G_{t+1} - T_{t+1},$$

  where B_t is end-of-period t nominal debt, $R_{t+1} \equiv 1 + i_{t+1}$ is the gross nominal return on government debt in period $t+1$, and $G_{t+1} - T_{t+1}$ is the nominal primary deficit in period $t+1$.
- The budget constraint is written in nominal terms. The debt-to-GDP ratio $b_t = B_t/Y_t$ is unit-free (inflation cancels in the ratio).
- Denote the primary deficit $PD_{t+1} \equiv G_{t+1} - T_{t+1}$ and the primary surplus $S_{t+1} \equiv T_{t+1} - G_{t+1} = -PD_{t+1}$.
- We abstract from money growth (we can treat money as a type of government debt).
- Note: is it really a "constraint"?

DEBT TO OUTPUT RATIO

- Define the gross nominal growth rate $\Gamma_{t+1} \equiv Y_{t+1}/Y_t$. Divide by Y_{t+1}:

$$b_{t+1} = \frac{R_{t+1}}{\Gamma_{t+1}} b_t + pd_{t+1},$$

  where $b_t \equiv B_t/Y_t$ and $pd_t \equiv (G_t - T_t)/Y_t$.

- Define cumulative objects from period s to period t:

$$\mathcal{G}_{s:t} \equiv \frac{Y_t}{Y_s}, \qquad \mathcal{R}_{s:t} \equiv \prod_{k=s+1}^{t} R_k.$$

  $\mathcal{G}_{s:t}$ is how much GDP grew; $\mathcal{R}_{s:t}$ is how much \$1 of debt grew (by compounding returns in periods $s+1, \ldots, t$).

DEBT TO OUTPUT RATIO

- For simplicity assume $B_0 = 0$.
- Iterating backward:

$$b_{t+1} = \sum_{s=1}^{t+1} \frac{\mathcal{R}_{s:t+1}}{\mathcal{G}_{s:t+1}} pd_s.$$

- Debt to output ratio today is determined by:
  1. The past primary deficits to GDP ratios;
  2. The past returns on debt;
  3. The past growth rates of (nominal) GDP.
- We can use the formula above to understand what determines the current debt to output ratio.

CLASSIC DEBT SUSTAINABILITY ANALYSIS

- The backward formula is pure accounting: it decomposes how past deficits, returns, and growth produced today's debt. It is an identity – always true, for any b_t.
- Warning: sustainability is a forward-looking question – can today's debt be repaid from *future* surpluses? The backward decomposition cannot answer this (D'Erasmo et al. 2016).
- Classic DSA sidesteps this by assuming a deterministic steady state with one-period debt and asking: what debt ratio is consistent with a given long-run primary balance?

CLASSIC DSA: THE DIFFERENCE EQUATION

- Assume constant $\bar{\Gamma}$, $\bar{R}$, and define the primary surplus $s \equiv -pd$. The debt ratio evolves as:

$$b_{t+1} = \frac{\bar{R}}{\bar{\Gamma}} b_t - s.$$

- Equivalently, the change in debt is

$$b_{t+1} - b_t = \underbrace{\left(\frac{\bar{R}}{\bar{\Gamma}} - 1\right) b_t}_{\text{interest–growth differential}} - s.$$

The first term is the interest–growth gap applied to current debt; the second is the surplus that pays it down.

CLASSIC DSA: THE STEADY STATE

- Setting $b_{t+1} = b_t$ in $b_{t+1} = (\bar{R}/\bar{\Gamma})\, b_t - s$:

$$b^{ss} = \frac{s\,\bar{\Gamma}}{\bar{R} - \bar{\Gamma}}.$$

- This requires $\bar{R} \neq \bar{\Gamma}$. The sign of $\bar{R} - \bar{\Gamma}$ determines everything:
  - $\bar{R} > \bar{\Gamma}$: a positive surplus $s > 0$ is needed for $b^{ss} > 0$.
  - $\bar{R} < \bar{\Gamma}$: even a deficit ($s < 0$) yields a positive, stable steady-state debt.

CLASSIC DSA: STABILIZING THE DEBT RATIO

- What surplus s^* keeps the debt ratio constant at its current level b?
- Setting $b_{t+1} = b_t = b$:

$$s^* = \frac{\bar{R} - \bar{\Gamma}}{\bar{\Gamma}} b.$$

  - $\bar{R} > \bar{\Gamma}$: must run a surplus proportional to debt – the "debt burden".
  - $\bar{R} < \bar{\Gamma}$: can run a deficit and still keep debt stable – growth does the work.
  - The required adjustment is larger the higher is b: high debt requires large surpluses.
- This formula is the workhorse of IMF and ECB debt sustainability assessments.

CLASSIC DSA: DEBT DYNAMICS

- Suppose the government runs a constant surplus s every period. Starting from b_0:

$$b_{t+1} = \left(\frac{\bar{R}}{\bar{\Gamma}}\right)^{t+1}\left(b_0 - b^{ss}\right) + b^{ss}.$$

- $\bar{R} > \bar{\Gamma}$: $(\bar{R}/\bar{\Gamma})^t \to \infty$ – debt explodes unless $s \geq s^*$. As debt rises, the required surplus rises with it – the snowball effect.
- $\bar{R} < \bar{\Gamma}$: $(\bar{R}/\bar{\Gamma})^t \to 0$ – debt converges to b^{ss} from any starting point. Growth erodes debt automatically.
- Key takeaway: $\bar{R} \gtrless \bar{\Gamma}$ determines whether debt dynamics are explosive or self-correcting.

THE r VERSUS g DEBATE

- If $\bar{\Gamma} > \bar{R}$ it seems there is no fiscal cost to debt.
- Note: $\bar{R}/\bar{\Gamma} = (1+r)/(1+g)$ where r and g are real – inflation cancels. So $\bar{\Gamma} > \bar{R} \Leftrightarrow g > r$: the "$r$ versus g" debate.
- Blanchard (2019) argues that $g > r$ is the norm, not the exception.
- But what is r? How to measure it? Blanchard (2019) examines the 1-year Treasury bill rate, the 10-year bond rate, adjusts for various maturities…
- Note: $g > r$ does not mean it is optimal to run deficits – it means debt is self-correcting, not that it is free.

APPLICATION: MAASTRICHT CRITERIA

- The Treaty of Maastricht set $b \leq 60\%$ and the total deficit ratio $d \leq 3\%$.
- Caution: d includes interest payments, unlike the primary deficit pd.
- Define $d_{t+1} \equiv (G_{t+1} + i_{t+1} B_t - T_{t+1})/Y_{t+1}$. Since $B_{t+1} = B_t + D_{t+1}$:

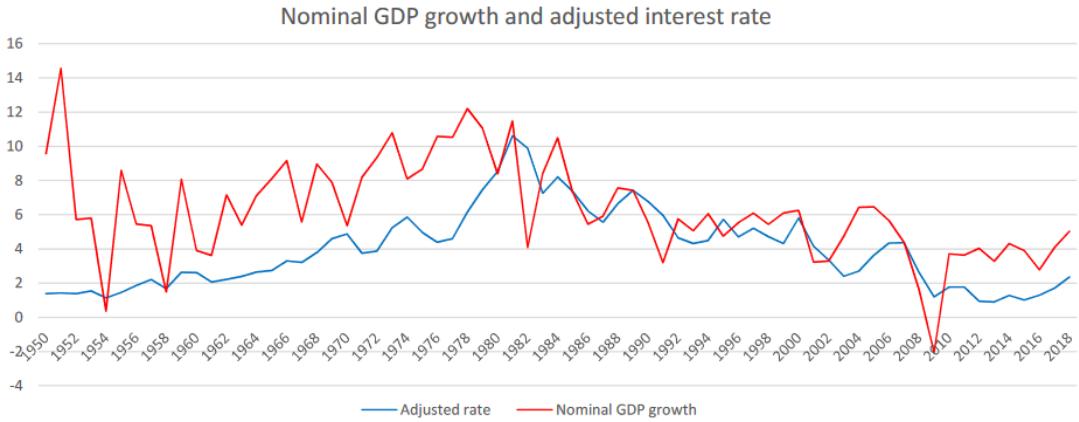
$$b_{t+1} = \frac{b_t}{\Gamma_{t+1}} + d_{t+1}.$$

- In steady state with nominal GDP growth $x \equiv \bar{\Gamma} - 1$:

$$b \approx \frac{\bar{d}}{x}.$$

- For $b = 0.6$ and $\bar{d} = 0.03$: $x = 5\%$ nominal GDP growth. In the 1990s EU this meant $\approx 3\%$ real growth + 2% inflation – plausible then, increasingly unrealistic since.
- Note: $\bar{R}$ does not appear – the interest cost is already embedded in $\bar{d}$.

RATES IN THE US



Source: Blanchard (2019)

WHY DOES $r < g$ MATTER? DIAMOND (1965)

- If $r < g$ (equivalently $\bar{R} < \bar{\Gamma}$), the economy may be dynamically inefficient (Diamond 1965).
- Dynamic inefficiency: capital stock is too large; the economy overaccumulates. Reducing investment raises consumption for *every* generation.
- Government debt crowds out capital – but here crowding out is welfare-improving: debt absorbs excess savings and moves the economy toward the Golden Rule.

BLANCHARD (2019): WELFARE EFFECTS OF DEBT ARE AMBIGUOUS

- Fiscal cost of debt is zero when $\bar{R} < \bar{\Gamma}$: debt rollovers are feasible.
- But what about welfare? Two channels work in opposite directions:
  1. Safe rate channel: if the real safe rate $< g$, an intergenerational transfer via debt raises welfare.
  2. Risky rate channel: if the real marginal product of capital $> g$ (which it is empirically), crowding out capital lowers welfare.
- The net effect is ambiguous: debt may improve welfare (if the safe rate channel dominates) or may reduce it (if the capital channel dominates). Either sign is possible.

IS THE US DYNAMICALLY INEFFICIENT? ABEL ET AL. (1989)

- The real safe rate being below g does not imply dynamic inefficiency.
- Abel et al. (1989): the relevant test compares gross capital income (profits, rents, interest) to gross investment.
- If capital income exceeds investment, the economy is dynamically efficient – capital is not overaccumulated.
- The US passes this test: the economy is dynamically efficient despite the real safe rate being below g.
- Key distinction: the real safe rate can be below g while the real marginal product of capital is above g.

A GOLDILOCKS THEORY OF FISCAL DEFICITS (MIAN, STRAUB, SUFI 2022)

- Blanchard (2019): $\bar{R} < \bar{\Gamma} \Rightarrow$ deficit is free. Not so fast.
- Mian, Straub, Sufi (2022): $\bar{R} < \bar{\Gamma}$ arises from a convenience yield on government bonds – investors value their safety and liquidity, so they accept a return below the growth rate.
- Inequality amplifies this: when the rich save more (high wealth concentration), the convenience yield rises, pushing R further below Γ.
- But as the government issues more debt, the convenience yield shrinks (bonds become less scarce), pushing R back toward Γ. This tightens the government's own budget constraint.
- There is a Goldilocks zone: a moderate deficit that
  1. exploits the convenience yield (keeps $R < \Gamma$);
  2. does not push debt so high that the convenience yield vanishes and R catches up with Γ.

A GOLDILOCKS THEORY OF FISCAL DEFICITS (CONT.)

- Key insight: a "free lunch" (permanent deficit with no future taxes) requires $R < G - \varphi$, where φ is the sensitivity of $R - G$ to debt. This is stricter than $R < G$.
- The deficit–debt locus is hump-shaped. Small permanent deficits are sustainable; large ones cause debt to explode because R rises endogenously toward G.
- Contrast with Diamond (1965): Diamond's welfare gain from debt relies on crowding out *excess capital*. MSS show the same $R < G$ can arise from convenience yields with no overaccumulation – the welfare arithmetic is different.

CALIBRATED GOLDILOCKS ZONE (MSS 2022)

- Calibrated to December 2019 (pre-Covid), with $\varphi \approx 1.7\%$ (range: 1.1–2.5%).
- United States: max permanent deficit $z^* = 2.0\%$ of GDP at $b^* = 109\%$. The $R = G$ threshold is at $\bar{b} = 218\%$. The US was just inside the free lunch zone.
- Japan: max permanent deficit $z^* = 3.5\%$ of GDP at $b^* = 223\%$. The $R = G$ threshold is at $\bar{b} = 446\%$. Japan was in the inverted ZLB regime: raising the deficit could *reduce* debt (via higher inflation and nominal growth).
- Key takeaway: even with $R < G$, there is a finite Goldilocks zone. Exceeding it means deficits must eventually be repaid.

CLASSIC DSA: WHAT IT CANNOT TELL US

- It only defines what long-run debt is for a given long-run primary balance (or vice versa) if stationarity holds, or defines lower bounds on the short-run dynamics of the primary balance.
- It does not pin down which steady state b^{SS} the economy should converge to – there is no optimality criterion built in.
- For a given initial b_{t-1}, there are multiple paths of the primary balance consistent with government solvency – with different values of b^{SS} and different transitional dynamics.
- Some of these paths converge to a finite steady state; others let debt diverge to infinity – yet all are consistent with the government repaying its debt! We formalize this next.

CLASSIC DSA: WHAT IT CANNOT TELL US (CONT.)

- The correct solvency condition (derived next): the value of debt must equal the present discounted value of future primary surpluses.
- Wrong discount rate: discounting at the risk-free rate $\bar{R}$ is appropriate only under perfect foresight or risk-neutrality. If surpluses are procyclical (fall in recessions), investors demand a risk premium – discounting at $\bar{R}$ overstates the present value of surpluses and understates the fiscal burden.
- Cannot rank paths: even when multiple fiscal paths are all consistent with solvency, they have very different implications for output, prices, and welfare. Classic DSA cannot evaluate which path is better – it only checks whether the constraint is met, not what the adjustment costs.

INTERTEMPORAL GOVERNMENT BUDGET CONSTRAINT

IGBC

- We used the government budget constraint by solving it backward (go back in time).
- We can also solve it forward – the valuation approach, the market value of government debt is determined by the discounted value of future government surpluses.
- This idea is often used in finance (e.g., Campbell and Shiller 1988) - what determines stock prices?
- Allows us to think seriously about risk and asset pricing.

IGBC

- Ignore risk for now and rewrite $B_{t+1} = R_{t+1} B_t + PD_{t+1}$ as

$$B_t = \frac{1}{R_{t+1}} \left(B_{t+1} + S_{t+1} \right).$$

- Continue this process to get

$$B_t = \sum_{j=1}^{T} \frac{S_{t+j}}{\mathcal{R}_{t:t+j}} + \frac{B_{t+T}}{\mathcal{R}_{t:t+T}}.$$

IGBC

- Note: there is a term $B_{t+T}/\mathcal{R}_{t:t+T}$.
- If this term does not vanish as $T \to \infty$, there is a bubble.
- Imposing the no-Ponzi game condition ($\lim_{T\to\infty} B_{t+T}/\mathcal{R}_{t:t+T} = 0$) we get the IGBC:

$$B_t = \sum_{j=1}^{\infty} \frac{S_{t+j}}{\mathcal{R}_{t:t+j}}.$$

- If the above is satisfied, we say that IGBC holds.

IGBC

- The government only faces the period budget constraint – that is pure accounting.
- The IGBC emerges when bond markets clear: investors are willing to hold government debt at positive prices.
- Analogy: the IGBC is a valuation equation, like the stock price formula. Government bonds are a claim on future surpluses, just as stocks are a claim on future dividends (Jiang et al. 2024).
- The no-Ponzi condition rules out bubbles in government debt. It is enforced by bondholders: investors buy bonds only if they expect repayment. If they lose confidence $\Rightarrow$ bond prices fall $\Rightarrow$ yields spike $\Rightarrow$ the government loses market access.

WHEN THE IGBC FAILS: DEFAULT

- If no feasible sequence of surpluses can satisfy the IGBC at current prices, the government must default (explicitly or via inflation).
- There is a political limit to how large a surplus the government can run. If debt is so high that even the maximum feasible surplus cannot cover the interest–growth gap, the IGBC breaks down.
- Ghosh et al. (2013) call this fiscal fatigue: at high debt levels, the surplus response weakens. This implies a debt limit beyond which stabilization is impossible.
- We will formalize this idea when we discuss fiscal reaction functions (Bohn 1998, 2008).

IGBC WITH RISK

- Under uncertainty the IGBC becomes

$$B_t = \mathbb{E}_t \sum_{j=1}^{\infty} M_{t,t+j} S_{t+j},$$

  where $M_{t,t+j}$ is the nominal stochastic discount factor (SDF) – it prices nominal payoffs – and $S_{t+j} = T_{t+j} - G_{t+j}$ is the nominal primary surplus.
- $M_{t,t+j}$ reflects how holders of government debt discount future cash flows.
- It is a function of the state of the economy at t and $t + j$. Recall the first-order condition from the household problem.

RISK AND THE DISCOUNT RATE

- Define the nominal risk-free cumulative discount factor $\mathcal{R}^f_{t:t+j} \equiv 1/\mathbb{E}_t[M_{t,t+j}]$.
- By the covariance decomposition:

$$\mathbb{E}_t[M_{t,t+j}\, S_{t+j}] = \frac{\mathbb{E}_t\ S_{t+j}}{\mathcal{R}^f_{t:t+j}} + \mathrm{Cov}_t\big(M_{t,t+j},\, S_{t+j}\big)\,.$$

- Hence discounting surpluses at the risk-free rate is valid only if $\mathrm{Cov}_t(M, S) = 0$.

RISK AND THE DISCOUNT RATE (CONT.)

- If surpluses are procyclical (S_{t+j} falls in bad times when $M_{t,t+j}$ is high): $\text{Cov}_t(M, S) < 0 \Rightarrow$ discounting at $\mathcal{R}^f$ overstates the PDV of surpluses.
- Implication: Classic DSA and the deterministic IGBC are not reliable guides to true fiscal space.
- Next lecture: how to measure the correct SDF and risk-adjusted fiscal space (asset-pricing approach to debt sustainability).

DEBT SUSTAINABILITY

- We can say that debt is sustainable if and only if the IGBC holds.
- Problem: this condition is about the entire future.
- Solution (?): use forecasts of future s_t to compute the present value of future surpluses. Some early papers did this, but they used risk-free rates.
- Valid if one of these conditions holds:
  1. There is perfect foresight;
  2. Investors are risk-neutral;
  3. Primary surpluses do not covary with the SDF.

BOHN (1998)

- Not even $b_t \to \infty$ means that the IGBC does not hold – debt has to go to infinity slowly enough.
- Bohn (1998): see if the government does something that guarantees the IGBC holds, investigate the fiscal reaction function.
- Allows us to sidestep the problem of forecasting future surpluses and choosing the correct discount rate.
- Sufficient condition: the IGBC might hold even if Bohn's condition is not satisfied, but if the condition *is* satisfied, the IGBC holds for sure.

BOHN (1998)

- Linear fiscal reaction function:

$$s_t = \rho\, b_{t-1} + z_t + \epsilon_t.$$

- s_t is the primary surplus ratio; z_t captures other determinants of the surplus.
- Check if $\rho > 0$ – the government raises the surplus when debt is high.
- If $\rho > 0$, then the IGBC holds even if ρ is below $\bar{R}/\bar{\Gamma} - 1$.

BOHN (1998)

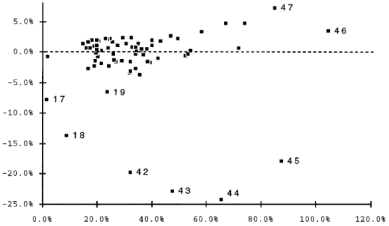
- If $\rho > 0$, the IGBC holds for any initial level of debt.
- This analysis works also for $\bar{R} = \bar{\Gamma}$ – there was division by zero in the classic analysis.
- If $\bar{R}/\bar{\Gamma} - 1 > \rho > 0$, debt explodes, but the IGBC still holds (under certain conditions: see Bohn 2007).

BOHN (1998)

- Bohn (1998) estimates ρ for the US in 1916–1995.
- He includes the level of temporary government spending and a business cycle indicator in z_t.
- He finds a positive value of ρ, around 0.05 for the entire sample.

BOHN (1998)

(a) The simple correlation



(b) With adjustment for temporary spending and output fluctuations

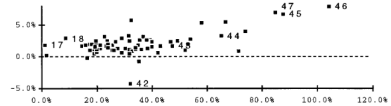


FIGURE I

Primary Surplus versus Initial Debt

The graph shows the privately held government debt/GDP at the start of a period on the horizontal axis against the primary budget surplus/GDP on the vertical axis, for 1916–1995; (a) shows raw data, and (b) shows the adjusted surplus, as explained in the text.

Source: Bohn (1998)

BOHN (1998)

TABLE I
DETERMINANTS OF THE BUDGET SURPLUS

Dependent variable primary budget surplus divided by GDP (s_t)							
Sample	Constant	*G*VAR	*Y*VAR	d_t	R^2	σ	*DW*
(1) 1916–1995	−0.019	−0.776	−1.450	0.054	0.936	0.014	1.42
	(−5.424)	(−33.001)	(−3.628)	(6.048)			
	[−3.957]	[−20.874]	[−4.075]	[3.787]			
(2) 1920–1995 excl. 1940–1947	−0.009	−0.551	−1.906	0.028	0.618	0.011	1.40
	(−2.030)	(−4.034)	(−4.666)	(2.701)			
	[−2.155]	[−3.721]	[−4.296]	[2.491]			
(3) 1916–1983	−0.018	−0.782	−1.414	0.054	0.942	0.014	1.54
	(−4.903)	(−31.667)	(−3.360)	(5.996)			
	[−3.958]	[−20.943]	[−4.004]	[4.076]			
(4) 1920–1982 excl. 1940–1947	−0.008	−0.520	−1.912	0.030	0.630	0.011	1.56
	(−1.710)	(−3.612)	(−4.441)	(2.815)			
	[−1.932]	[−3.272]	[−3.959]	[2.856]			
(5) 1948–1995	−0.015	−0.593	−2.139	0.037	0.651	0.010	1.54
	(−3.536)	(−4.182)	(−4.361)	(3.589)			
	[−3.496]	[−3.701]	[−3.757]	[2.821]			
(6) 1960–1984	−0.013	−0.410	−2.051	0.044	0.724	0.007	1.43
	(−2.110)	(−2.173)	(−4.174)	(2.028)			
	[−2.174]	[−2.281]	[−3.391]	[2.587]			

The variable d_t is the privately held debt/GDP at the start of the year. *G*VAR and *Y*VAR are measures of temporary government spending and of cyclical variations in output, respectively, from Barro [1986a]. All estimates are OLS with annual data; ( ) = ordinary *t*-statistics; [ ] = heteroskedasticity- and autocorrelation-consistent *t*-statistics (computed with Newey-West lag window of size 1); σ = standard error; *DW* = Durbin-Watson statistic.

Source: Bohn (1998)

FISCAL REACTION FUNCTIONS

- Bohn (2008) extends the analysis to 1792–2003. He finds $\rho > 0.1$, more than twice as large as in the earlier study.
- Mendoza and Ostry (2008) study fiscal reaction functions for a panel of multiple countries – similar results.
- Ghosh et al. (2013) show that ρ is much lower at high levels of debt – evidence of fiscal fatigue.
- But does the FRF remain stable over time? D'Erasmo et al. (2016) investigate.

MULTIPLE PATHS, SAME SOLVENCY

- Combine $b_t = (\bar{R}/\bar{\Gamma})\, b_{t-1} - s_t$ with the linear FRF $s_t = \mu_t + \rho\, b_{t-1} + \varepsilon_t$:

$$b_t = -\mu_t + \left(\frac{\bar{R}}{\bar{\Gamma}} - \rho\right) b_{t-1} - \varepsilon_t.$$

- If μ_t and ε_t are stationary, then b_t is stationary only if $\rho > \bar{R}/\bar{\Gamma} - 1$. But the IGBC holds for any $\rho > 0$: debt grows slower than $\bar{R}/\bar{\Gamma}$, so the discounted value vanishes.
- Implication: halving ρ or lowering the intercept μ yields a *different* long-run debt ratio and different transitional dynamics, but the *same* present value of surpluses. The classic DSA cannot distinguish between these paths.

STRUCTURAL INSTABILITY OF THE FRF

- D'Erasmo, Mendoza, and Zhang (2016) re-estimate Bohn's FRF for the U.S. using data from 1791 to 2014.
- Pre-2008 sample: $\hat{\rho} \approx 0.10$, consistent with Bohn (2008).
- Full sample including 2008–2014: $\hat{\rho}$ drops to ≈ 0.08. The Chow test rejects parameter stability.
- FRF residuals for 2009–2011 are twice the minus-two-standard-error bound. The FRF predicted rising primary surpluses; the data showed persistent, large deficits.
- This pattern of prolonged primary deficits after a debt surge is unprecedented in U.S. fiscal history – after the Civil War, WWI, and WWII, surpluses appeared within three years.

FISCAL REACTION FUNCTIONS

- Leeper and Li (2017) warn against using surplus-debt regressions to assess debt sustainability.
- For the estimator of ρ to be consistent, we must have

$$\mathbb{E}(\epsilon_t \mid b_{t-1}) = 0.$$

  1. This means that shocks at $t-1$ that affect b_{t-1} must not affect ϵ_t.
  2. This means that b_{t-1} cannot depend on the expectation of ϵ_t.
- Since the value of debt depends on the expected value of future surpluses, this is a strong assumption: ϵ_t could be serially correlated.

TAKEAWAYS

- Classic DSA: whether debt is sustainable depends on $\bar{R}$ versus $\bar{\Gamma}$ and the primary balance – but the framework is silent on risk and cannot rank fiscal paths.
- The IGBC (forward-looking valuation) is the right framework, but checking it requires the correct stochastic discount factor, not the risk-free rate.
- Bohn's fiscal reaction function offers an elegant shortcut: if $\rho > 0$, the IGBC holds. But the FRF may be structurally unstable and the estimator may be biased.
- Next lecture: the asset-pricing approach to debt sustainability – how to measure the correct SDF and risk-adjusted fiscal space.